

System Programming in C – Homework Exercise 1

Date published: *Thursday, March 19, 2026*
Date assigned: *Sunday, March 22, 2026*
Date due: ~~*Sunday, March 29, 2026 @ 21:00*~~
Sunday, April 5, 2026 @ 21:00

Preliminary preparations: Log into the course server (`sysprog.runi.ac.il`) with your user id and password, create the exercise directory `exercises/ex1/`, and enter it.

If you haven't already changed your password, then do so by using the command `passwd`.

General guidelines: read carefully before proceeding!!

- Do all your work in the directory `~/exercises/ex1/` that you create for this assignment.
- Problem 1 specifies a list of nine file manipulation tasks for you to implement.
- Each task should be implemented using a single Linux command.
- Use only relative paths; do not use absolute paths. Usage of absolute paths will cause the testing scripts to fail. The only exception to this rule is when you access the shared data directory `/share/ex_data/ex1/` in task #1.
Tip: if your path starts with `/` or `~`, then it is absolute.
- In Problem 2 you are required to type these commands into a solution file (`my_commands.txt`), which is the file that you submit.
- You should test your solution by following the four validation steps listed on page 3. Your script should pass the final validation step (running `test_ex1`) to guarantee correct submission.
- Follow the submission instructions on page 4. See **Homework submission instructions** for important general details on the submission process for homework assignments in the course.

Problem 1:

1. Copy the directory `/share/ex_data/ex1/bank_statements` into the current directory. After doing this, your current directory should contain the `bank_statements` directory, and that directory should contain two files (`hapoalim.xlsx` `mizrahi`) and one subdirectory (`leumi`), which contains 3 files (`January.pdf` `February.pdf` `Mar.pdf`)
2. Move into the `leumi` subdirectory (inside the newly copied directory `bank_statements`).
3. Create a copy of the file `January.pdf` and name this file `April.pdf`.
4. Change the name of file `Mar.pdf` to `March.pdf`.
5. Create a copy of the current directory (`leumi`) named `leumi_old_account` in the parent directory `bank_statements`. After this operation, the parent directory `bank_statements` should contain two subdirectories named `leumi` and `leumi_old_account` with identical contents (files `January.pdf` `February.pdf` `March.pdf` `April.pdf`).
6. Move to the parent directory (`bank_statements`).
7. Change the permissions of all files in directory `leumi` whose name contains `ary` so that their new permission code will be `-rwxr-----`. Use an octal permission code for this, and use the appropriate wildcard expression that does not make any assumption about the names of files in `leumi`.
8. Delete all files from directory `leumi_old_account` (without deleting the directory itself). Again, use the appropriate wildcard expression that does not make any assumption about the names of files in `leumi_old_account`.
9. Use `echo` to append the line of text `get updated info` to the end of file `mizrahi`.
10. Add write permissions for users in your group (students) to your current directory (`bank_statements`) and all items contained in it (recursively). Use an alphabetic permission specifier for this.

Problem 2:

Use a text editor (such as `pico`) to type the commands you used in Problem 1 into a text file named `my_commands.txt`. The file should contain exactly 10 lines, each containing a single command. This allows the testing script to test your solution and give you feedback for possible corrections. Make sure that each of the 10 lines in the file contains only the executed command and no additional documentation and labels (which may cause an error while running).

Validation and testing:

1. Verify that file `my_commands.txt` has exactly 10 lines by using command `wc`:
`==> wc -l my_commands.txt`

You will know that you have exactly 10 lines if the result is:

```
10 my_commands.txt
```

2. Execute the commands in file `my_commands.txt` as a script using command `source` and make sure that **no error messages** are printed:

```
==> source my_commands.txt
```

```
==>
```

3. examine the contents of the resulting directory structure (using command `ls`):

```
==> ls -l
```

```
total <SOME NUMBER HERE>
```

```
-rw-rw-r-- 1 <USER ID> students      9 <TIME STAMP> hapoalim.xlsx
drwxrwxr-x 2 <USER ID> students 4096 <TIME STAMP> leumi
drwxrwxr-x 2 <USER ID> students 4096 <TIME STAMP> leumi_old_account
-rw-rw-r-- 1 <USER ID> students    25 <TIME STAMP> mizrahi
```

```
==> ls -l leumi
```

```
total <SOME NUMBER HERE>
```

```
-rw-rw-r-- 1 <USER ID> students      8 <TIME STAMP> April.pdf
-rwxrw---- 1 <USER ID> students      9 <TIME STAMP> February.pdf
-rwxrw---- 1 <USER ID> students      8 <TIME STAMP> January.pdf
-rw-rw-r-- 1 <USER ID> students      6 <TIME STAMP> March.pdf
```

```
==> ls -l leumi_old_account
```

```
total 0
```

4. Make sure that the permission string of each item is as specified above.
5. Examine the contents of the file `mizrahi` (using `cat`):
`==> cat mizrahi`
`mizrahi`
`get updated info`
6. Run your script from **different locations** (not just `~/exercises/ex1/`) to ensure that it does not contain any unwanted absolute paths.
7. Use the script `/share/ex_data/ex1/test_ex1` to test your solution. Execute the following command from directory `~/exercises/ex1/`:

```
==> /share/ex_data/ex1/test_ex1
```

(the script produces a detailed error report to help you debug your code)

Submission Instructions:

1. After you validated and tested your solution, make sure that your `~/exercises/ex1/` directory contains the `my_commands.txt` script and a `PARTNER` file containing the user id of the non-submitting partner.
2. Only one partner should submit a solution file (`my_commands.txt`). The non-submitting partner should only submit a `PARTNER` file containing the user id of the submitting partner.
3. If you are submitting alone, specify your own user id in the `PARTNER` file.
4. Check your entire submission (solution file and `PARTNER` file) by running `check_ex ex1`. The script should be executed from the account of the submitting partner, and it may be run from any directory. The script will provide a detailed report, including the report produced by the test script `/share/ex_data/ex1/test_ex1`. **Clean execution of this script guarantees you 80% of the assignment's grade.**
5. Once you are satisfied with your solution, you may submit it by running `submit_ex ex1`. The script should be executed from the account of the submitting partner, and it may be from any directory. The submission script creates a read-only directory `~/exercises/ex1/submit/` containing your solution file and the report from script `check_ex`. You may modify your submission any time before the deadline by running `submit_ex ex1 -o` from any location.
6. For more information on the submission process, see the [Homework submission instructions](#) file on the course website.